SHREE VENKATESHWARA HI-TECH ENGINEERING COLLEGE

TEAM NAME - BIOPULSE
THEME - DIAGNOSTICS & MEDICAL DEVICES
TRACK CODE - BB-DM-01

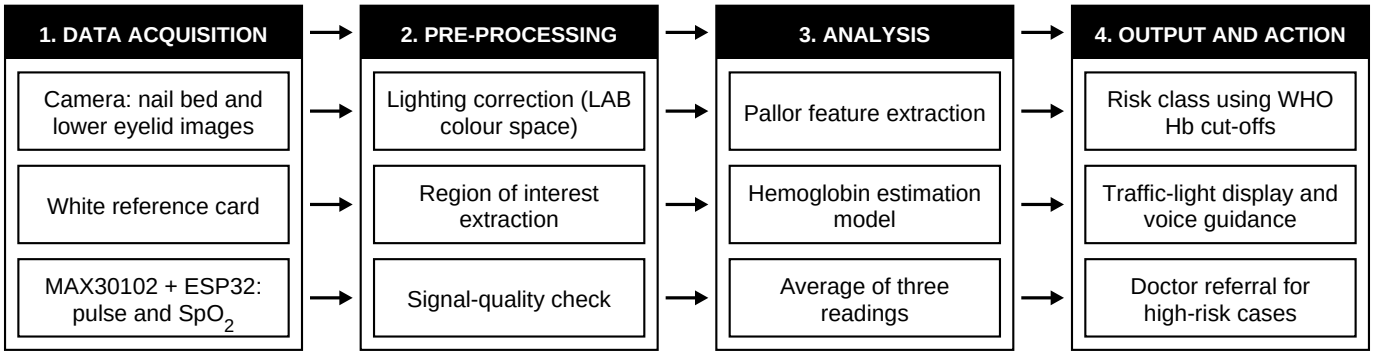
PROBLEM STATEMENT

Non-invasive anemia screening that works at an Anganwadi without a blood drop or laboratory support.

PROPOSED SOLUTION

Our solution is an offline, non-invasive anemia screening kit. A camera captures the finger nails and lower eyelids with a white colour-reference card; OpenCV corrects the lighting and extracts pallor features, and a trained model estimates hemoglobin and anemia risk. A MAX30102 sensor on an ESP32 records pulse and SpO_2 to check signal quality and flag non-anemia causes; it supports the camera result and does not estimate hemoglobin alone. Three readings are averaged and shown instantly as a traffic-light result (Low / Moderate / High) with local-language voice guidance. High-risk cases are referred for a confirmatory blood test.

SYSTEM ARCHITECTURE



OBJECTIVES

- Detect anemia risk without a blood test.
- Improve accuracy with OpenCV colour correction and a trained model.
- Take three readings and show the average result.
- Give quick results; recommend a doctor if needed.
- Work offline at low cost for Anganwadi workers.

KEY FEATURES

- Nail and eyelid analysis with a reference card.
- MAX30102 pulse, SpO_2 and signal-quality check.
- Three-level risk using WHO Hb cut-offs.
- Traffic-light screen with local-language voice.
- Offline web app with doctor-referral alert.

CATEGORY | TOOLS

Category	Tools
Hardware	MAX30102 sensor, ESP32, camera, white reference card, LED lighting shroud
Software	Python, OpenCV / OpenCV.js, HTML, CSS, JavaScript (offline web app)
Model	Nail and eyelid colour features (LAB space) with regression / random forest